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Mastering Activity Relationships in Primavera P6: The Backbone of Dynamic Scheduling

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Meta Description:

Master the 4 types of activity relationships (FS, SS, FF, SF) in Primavera P6. Learn how Lag & Lead times impact Critical Path analysis and avoid common scheduling pitfalls in EPC projects.

Keywords:

Activity Relationships, Primavera P6 Scheduling, Finish to Start, Start to Start, Critical Path Method, Schedule Logic, Lag and Lead Time, Project Controls, EPC Planning, Network Diagram, Scheduling Best Practices.

Introduction: Why Relationships Matter More Than Dates

In professional project controls, a schedule without logic is merely a list of dates. The true power of the Critical Path Method (CPM) lies not in the duration of activities, but in the relationships that bind them together.

For planners and schedulers working on complex EPC projects, understanding activity relationships is non-negotiable. Incorrect logic leads to broken critical paths, unrealistic float calculations, and schedules that cannot withstand forensic delay analysis. This guide provides a definitive breakdown of relationship types in Primavera P6, best practices for application, and common pitfalls to avoid.

The Four Types of Activity Relationships

Primavera P6 supports four standard precedence relationships defined by the Project Management Institute (PMI). Understanding when to use each is the mark of a senior scheduler.

1. Finish-to-Start (FS) – The Industry Standard

Definition: Activity B cannot start until Activity A has finished.

Usage: This should represent 80-90% of all logic in your schedule. It represents natural sequential workflow (e.g., Pour Concrete → Cure Concrete).

Best Practice: Always default to FS unless there is a specific, documented reason to use another type. FS relationships produce the most predictable and defensible Critical Path.

2. Start-to-Start (SS) – Parallel Workflows

Definition: Activity B cannot start until Activity A has started.

Usage: Ideal for overlapping activities where work can proceed in parallel once a predecessor initiates (e.g., Excavation → Pipe Laying with a lag).

⚠ Caution: SS relationships often create "Open Ends" if not paired with a Finish constraint or successor. They can artificially inflate Total Float and hide true criticality.

3. Finish-to-Finish (FF) – Coordinated Completions

Definition: Activity B cannot finish until Activity A has finished.

Usage: Best for activities that must complete simultaneously or where one task depends on the completion of another for quality/compliance (e.g., System Installation → System Testing Documentation).

⚠ Caution: Like SS, FF relationships can mask critical path drivers. Use sparingly and always verify the resulting float values.

4. Start-to-Finish (SF) – The Rare Exception

Definition: Activity B cannot finish until Activity A has started.

Usage: Extremely rare in construction and engineering. Sometimes used in shift-work handovers or just-in-time material delivery scenarios.

⚠ Warning: Most scheduling standards (including DCMA 14-Point) flag SF relationships as potential errors. Avoid unless absolutely necessary and document the justification.

Lag and Lead Time: Fine-Tuning Your Logic

Relationships are rarely perfect zero-day links. Lag and Lead values adjust the timing between activities.

Term

Definition

Example

Best Practice

Positive Lag

Delay added after relationship trigger

FS + 5d (Curing time)

✅ Preferred. Represents real-world waiting periods.

Negative Lag (Lead)

Acceleration/overlap before trigger

FS - 3d (Early start)

⚠ Use cautiously. Can create illogical overlaps.

⚡ Critical Warning on Negative Lag

Negative lag is one of the most common causes of schedule rejection during baseline reviews. It implies that a successor starts before its predecessor's triggering event occurs. In forensic delay analysis, negative lag is frequently challenged as unrealistic.

Pro Tip: Instead of using negative lag, consider redefining the relationship type (e.g., changing FS-5d to SS+2d) or splitting the activity into two distinct phases with proper FS logic.

Relationships and the Critical Path

The type of relationship directly impacts Total Float and Critical Path identification:

FS Chains: Create clear, continuous critical paths. Float calculation is straightforward.

SS/FF Combinations: Can create "dangling logic" where an activity appears to have excessive float because only its start OR finish is constrained.

Multiple Predecessors: When an activity has both FS and SS predecessors, Primavera P6 calculates float based on the most restrictive relationship. Always review the "Predecessors" tab to understand which link is actually driving the date.

Driving vs. Non-Driving Relationships

In Primavera P6, check the "Driving" column in the Relationships tab:

Yes: This relationship determines the early/late dates of the successor.

No: This relationship exists but does not currently control timing (it has positive float relative to the successor).

Understanding driving relationships is essential for effective schedule compression and delay analysis.

Common Pitfalls & How to Avoid Them

Pitfall

Impact

Solution

Excessive SS/FF usage

Artificial float, hidden critical path

Convert to FS where possible; add missing successors

Negative Lag > 10% of duration

Unrealistic overlap, audit failure

Restructure logic or split activities

SF Relationships

Confusing logic, DCMA failure

Replace with FS/SS alternatives

Missing Relationships

Open ends, broken network

Run Schedule Log (F9) regularly

Hard Constraints overriding logic

Logic becomes irrelevant

Remove constraints; fix underlying relationships

DCMA 14-Point Assessment: Relationship Checks

When validating your schedule against DCMA 14-Point standards, pay special attention to:

Logic Density: Each activity should have at least one predecessor and one successor (except milestones).

Lag Usage: Total lag should not exceed 10% of the project duration.

Relationship Type Distribution: >80% should be FS. High percentages of SS/FF warrant investigation.

Orphaned Activities: Zero relationships = schedule integrity failure.

Conclusion: Logic First, Dates Second

A robust schedule is built on sound logic, not manipulated dates. By mastering activity relationships in Primavera P6, you ensure that your schedule is:

Dynamic: Responds accurately to progress updates and delays

Defensible: Withstands claims and forensic analysis

Actionable: Provides reliable critical path intelligence for decision-making

Remember: Every relationship tells a story about how the project will actually be built. Make sure that story is accurate, logical, and professionally structured.

About the Author

Behzad Mousavi is a Senior Project Controls Specialist and Primavera P6 Expert with extensive experience in EPC project planning, delay analysis, and schedule forensics. His articles provide practical, field-tested guidance for planning professionals worldwide.

For more technical articles, training materials, and professional resources, visit our page in GitHub as a link below:

- <https://sbmousavices.github.io/industrial-Tools>

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